

A DATA AND PROTECTION



1 Identity-disjoint data

MOBIO / LFW / FEI / SCface
Train / val / test people differ
Gallery never in exposures

2 Fixed face encoder

YuNet; 5-landmark alignment
ArcFace w600k_r50; 512-D
Unit norm; frozen backbone

3 Keyed protection

Template $T = f(\text{key}, x)$
Fresh / pool / shared key
Secret key values withheld

4 Exposure records

Input: batch $x \times n \times d$
8 nested sets per identity
 $n = 1, 10$ (initial: 1, 2, 5, 10)

OUTPUT d BioHash: 128 bits | MLP-Hash: 512 bits | IoM-GRP: 300 codes ($q=16$; one-hot $d=4,800$)
PolyProtect: 170 reals (window $m=5$, overlap=2). Scheme coverage differs by dataset.

B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



5 Supervised attacker

Single / mean / max MLP: $d \rightarrow 256 \rightarrow 512$
DeepSets: encode, mean, decode
Source embeddings: train targets only



6 Reconstructed embedding

L2-normalized prediction in 512-D
Train on non-test identities
Select checkpoint by validation loss



7 Held-out gallery linkage

Cosine scores vs. unprotected gallery
Top-1 / top-5 / AUROC / EER / TAR
Identity-clustered bootstrap intervals

KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

REPLICATION MOBIO / FEI follow-up: 2 identity partitions x 3 model seeds; matched 120-epoch training caps.

DESIGN Same-identity aggregation under controlled transform reuse; detailed attacker paths in the companion figure.